DAY 32 | POST-READ

# LLM Evaluation and Quality Gates: Recap and Beyond

What we covered today, plus ideas to stretch your understanding.

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## Part A: Summary of today

| Topic | What to remember |
|---|---|
| Golden dataset | Verified, representative, versioned, maintained. Fields: id, question, expected, relevant ids, tags, must include and must not include. |
| Prompt regression | Rerun the golden set for every prompt, model or retrieval change; compare per item. |
| Deterministic checks | Phrases, JSON validity, length. Fast; run on every commit. |
| Model-graded checks | Faithfulness, tone, correctness rubrics. Slower; nightly or before release. |
| Tool-call fidelity (NEW v4) | Right tool, schema-valid arguments, exact values, no call when not needed. |
| Context precision and recall | Relevant retrieved / retrieved; needed evidence found / needed evidence. |
| Citations | Every cited chunk was retrieved and is relevant. |
| BLEU | n-gram overlap with brevity penalty. Good for translation, weak for open answers. |
| Cost in the gate (NEW v4) | Tokens and estimated cost per query, compared with the baseline. |
| Eval-first with Codex (NEW v4) | Write the evals, see them fail, let Codex implement, keep evals as tests. |

## Part B: Going beyond today

### 1. Baselines from main

Compare a pull request's scores with the main branch's latest scores, not with fixed numbers only. A drop of more than a tolerance (for example 2 points) fails the gate even if the absolute score is still high.

### 2. Non-determinism

Model outputs vary between runs. Use temperature 0 where the model allows it, run important items several times, and set thresholds with a little slack rather than demanding perfection.

### 3. Cost-aware evaluation tiers

Run cheap deterministic and tool tests on every push, a small golden subset with model judges on pull requests, and the full suite nightly. Track the cost of the evaluation run itself.

### 4. Ready-made tools

Frameworks such as promptfoo, DeepEval, Ragas and the OpenAI Evals project help define datasets, metrics and CI integrations. Concepts from today carry over directly; check each tool's current documentation.

### 5. Keep the golden set alive

Add every production failure that users report as a new golden item, with an owner and date. The set grows to reflect reality.

## Part C: Check yourself

- What makes a dataset golden rather than just a test set?
- What four things does a tool-call fidelity test check?
- What is the difference between context precision and context recall?
- Why include cost per query in a quality gate?
- What does eval-first development with Codex look like, step by step?

## Part D: Practice ideas for tonight

- Add 10 golden items for your capstone domain with must include and must not include phrases.
- Add three tool-call fidelity cases, including one where no tool should be called.
- Wire the quality gate into your pipeline as a stage before deployment.

**Coming up on Day 33:** LLMOps: lifecycle, AgentOps, versioning, monitoring and cost strategies, plus GenAI assessment 3.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today